

Proposal: AI-Powered Email Assistant for Fintech Inbox Management

Executive Summary

We propose building a secure, AI-powered email automation agent tailored for the fintech industry. The system uses the latest Anthropic Claude models to read incoming emails, learn from your past email history and replies, draft accurate and compliant responses, and wait for human approval before sending anything.

Key benefits include:

- Faster email handling without losing control
- Consistent tone and regulatory compliance
- Continuous improvement from the real email data
- Full audit trail and analytics dashboard

This solution keeps a human in the loop at every step while significantly reducing manual work especially important in fintech where accuracy, security, and compliance are non-negotiable.

Project Overview & Objectives

Goal Create an autonomous yet fully supervised AI agent that manages inbox replies in a fintech environment.

Main Objectives

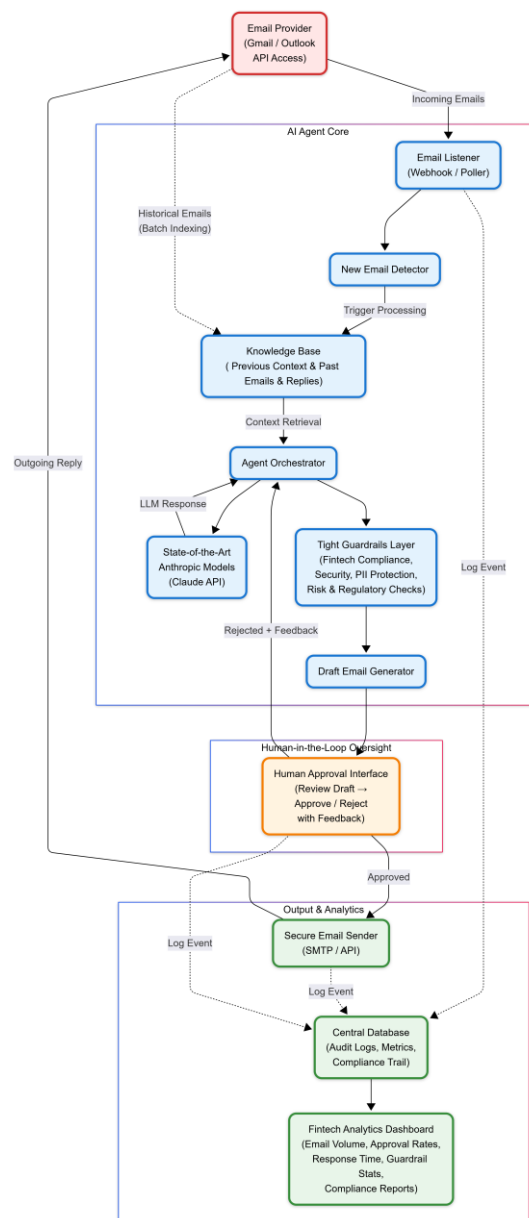
- Automatically detect and process new inbound emails
- Understand context using past emails and replies
- Generate high-quality draft replies powered by state-of-the-art Anthropic models
- Apply strict fintech-specific guardrails (compliance, PII masking, risk checks)
- Present drafts for human review and approval/rejection
- Send only approved emails
- Track performance and provide visibility through a dashboard

Scope Inclusions

- Integration with major email providers (Gmail, Outlook via APIs)
- Retrieval-Augmented Generation (RAG) for learning from historical emails
- Claude-powered drafting logic
- Multi-layer guardrails
- Human approval interface (web or email-based)
- Logging, analytics database, and dashboard

Proposed Solution – System Architecture

The architecture follows best practices for secure, observable AI agents in regulated industries. It separates concerns clearly: ingestion, intelligence, safety checks, human oversight, and analytics.



A clean flowchart showing:

- **Email Provider** (Gmail/Outlook) on the left
- Core AI Agent components in the middle (Listener → Detector → Knowledge Base/RAG → Orchestrator → Claude → Guardrails → Draft Generator)
- Human Approval Interface as a separate oversight layer
- Output side with Sender, Analytics Database, and Dashboard on the right
- Arrows showing incoming email flow, approval path, feedback loop, logging to analytics, and historical batch indexing for learning

The design ensures:

- No email is sent without explicit human approval
- All actions are logged for audit
- Sensitive data is protected at every step
- The system learns and improves over time without manual retraining

Key Components Explained

1. **Email Listener & Detector** Monitors inbox (via webhook or secure polling) and triggers processing only for relevant new messages.
2. **Knowledge Base** Stores vectorized past emails and replies so the AI can retrieve relevant context before drafting.
3. **Orchestrator + Anthropic Claude** Coordinates the flow and uses Claude (state-of-the-art LLM) to understand intent and generate natural, professional drafts.
4. **Tight Guardrails Layer** Fintech-specific checks including:
 - PII detection and masking
 - Regulatory keyword scanning (e.g. promises, guarantees)
 - Risk scoring of content
 - Block sending if confidence is too low
5. **Human Approval Interface** Clean dashboard/email notification showing:
 - Original email
 - AI draft
 - Confidence score
 - Approve / Reject / Edit buttons
 - Optional feedback to improve future drafts
6. **Secure Sender** Sends only approved emails using official API/SMTP.
7. **Analytics & Dashboard** Tracks:
 - Emails processed per day/week
 - Approval/rejection rates
 - Average response time
 - Guardrail triggers
 - Compliance audit logs

Technology Stack (High-Level)

- AI: Anthropic Claude API (latest models)
- Backend: Python
- Vector Database: Pinecone / Weaviate / PGVector (for RAG)
- Email Integration: Google Workspace API / Microsoft Graph API
- Frontend (Approval + Dashboard): React / Next.js
- Database (logs/metrics): PostgreSQL / MongoDB
- Hosting: AWS / Azure / GCP